

Where Classical ML Ends and Modern Deep Learning Begins

Choose tools from the problem—not from fashion

Broader Context · Foundations of Machine Learning



The Boundary Is Fuzzy

Classical ML

Linear/logistic regression, trees, SVMs, ensembles (random forests, gradient boosting), clustering, PCA — comparatively compact models, often with a handful to a few thousand parameters.

Deep learning

Composed neural networks — many layers of learned linear transformations plus nonlinearities — with learned internal representations and typically millions to billions of parameters.

Convention, not theorem

There is no formal threshold of "layers" or "parameters" that separates the two categories — the line is drawn by community usage, not mathematics.

$$\underbrace{\hat{y} = \sigma(w^\top x + b)}_{\text{logistic regression}} \iff \underbrace{\hat{y} = \sigma(W_2 \sigma(W_1 x + b_1) + b_2)}_{\text{one-hidden-layer neural network}}$$

A single-layer network with a sigmoid output *is* logistic regression. Add one hidden layer with a nonlinearity, and you already have a (shallow) neural network — the "classical" and "deep" labels are describing two ends of one continuous family, not two disconnected fields.

When Classical ML Is Often Better

Tabular data

Gradient-boosted trees (XGBoost, LightGBM) and random forests remain formidable, often winning defaults on structured spreadsheet-style data with heterogeneous feature types.

Small datasets

Lower-capacity methods need far fewer examples to generalize — recall the bias-variance tradeoff: a high-capacity deep network on a few hundred rows will overfit badly.

Interpretability requirements

Linear models and shallow trees can be read directly (recall the previous deck); this matters for regulated domains like lending and healthcare with legal explanation requirements.

Limited compute and iteration speed

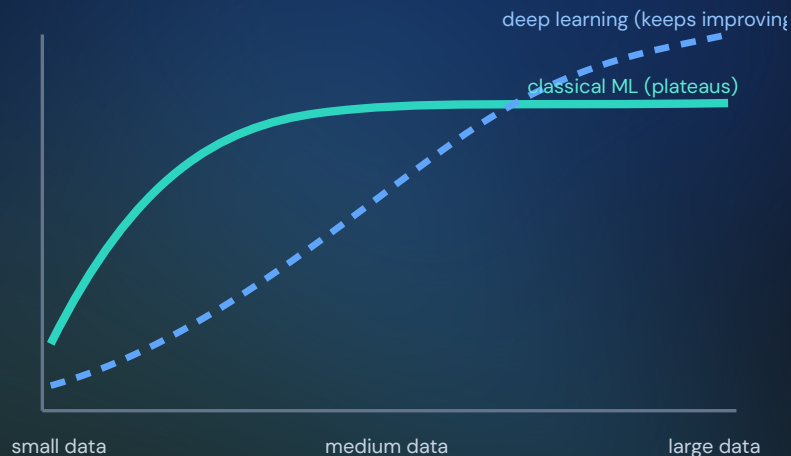
Training and serving classical models is usually orders of magnitude cheaper, letting you run more experiments per day of engineering time.

Where Deep Learning Changed the Landscape

Unstructured input — Images, audio, and text lack tabular structure; deep networks learn hierarchical features directly from raw pixels, waveforms, or tokens.

Favorable scaling — Larger networks trained on more data and compute keep improving well past the point where classical methods plateau.

Less manual feature engineering — Classical pipelines needed hand-designed features (edge detectors, n-grams); deep networks learn comparable features automatically.



A domain-dependent empirical pattern for unstructured data — not a universal law.

A Practical Choice Map

1. Data structure — Tabular (favor trees/ensembles) or unstructured — images, audio, text (favor deep learning)?

2. Data volume — Enough labeled examples to responsibly train a high-capacity model without overfitting?

3. Constraints — Interpretability requirements, latency budget, compute/serving cost, and who maintains this system after you leave the project?

1

Problem

2

Data + scale

3

Constraints

4

Model family

Worked example: 5,000-row spreadsheet of loan applications, regulator requires an explanation for every denial → tabular + small + high interpretability need → gradient-boosted trees or logistic regression, not a neural network.

The Mathematical Foundations Do Not Change

Linear algebra — Data, weights, and layer computations are still vectors, matrices, and matrix multiplications.

Probability and statistics — Uncertainty, likelihood-based losses (e.g., cross-entropy), and evaluation metrics carry over unchanged.

Calculus — Backpropagation is literally the multivariate chain rule applied layer by layer.

Methodology — Generalization, validation discipline, and this module's fairness/failure-mode habits apply to any model family.

1

Foundations

2

Classical ML

3

Deep learning

4

Future models

One Field, Many Useful Tools

Classical

Strong default for tabular data, small datasets, tight interpretability or compute constraints.

Deep

Strong on unstructured data (images, audio, text) at scale, where representation learning beats hand-engineered features.

Foundations

Linear algebra, probability, calculus, and validation methodology transfer to both, and to whatever comes next.

Next: the capstone applies the complete pipeline — modeling, evaluation, and ethical review — independently.